Midterm 02

ⓘ This is a preview of the published version of the quiz

Started: Oct 23 at 4:40pm

Quiz Instructions

This is Midterm 02. Once you start, you will have 1 hour and 30 minutes, but the test is designed to take less time. No partial credit is awarded, so be careful!

- Note that (unfortunately) the only fair way to proctor the exam is to do so without providing clarifications. If you feel that a question is ambiguous, please make a post to Campuswire after the exam and we will consider whether the grading of the question should be changed.
- If you have any technical issues during the exam (such as your internet going out) make a post to Campuswire. Don't worry: we'll figure something out.
- The exam is open book, open notes, open everything **except** you may not discuss the exam with another living organism until *after* the exam scores have been released.

Question 1 1 pts

Are you using Safari to take this exam? The correct answer is "no".

If you use Safari, you may be unable to see some of the images below. Apparently, Canvas is unable to work with some modern browsers...

- No
- Yes

Question 2 1 pts

Suppose you are given the list [12, 22, 99, 12, 101, 0, 12].

What is the **third** order statistic of this list?

Question 3

1 pts

Consider the problem of finding the kth order statistic in a **sorted** array of n numbers. Which of the below is a **tight** theoretical lower bound for this problem?

In other words, what is the best possible time complexity of any algorithm which solves this problem?

- Omega(n log n)
- Omega(n)
- Omega(log n)
- Omega(n^2)
- Omega(1)

Question 4

1 pts

Suppose you are given the input of [5, 10, -30, 42, 7, 3, 23]. You will be running quickselect on this input. Which of the below is the **worst** choice of pivot (among the choices given)?

- 7
- 42
- 5
- 23

Question 5

1 pts

Given a Python set (i.e., a hash table) of n numbers, and two numbers *a* and *b* (with *a* < *b)* what is the best possible average case time complexity of an algorithm which computes the number of elements in the set which are between *a* and *b*?

- Theta(log n)
- Theta(n)
- Theta(n^2)
- Theta(1)

Question 6

1 pts

A particular **directed** graph has seven nodes. What is the largest possible degree of any node in the graph?

Question 7

1 pts

A particular **undirected** graph has seven nodes. What is the largest possible degree of any node in the graph?

Question 8

1 pts

Let V = {a, b, c, d} and E = {(a,b), (a,a), (a,c), (d,b), (c,a)} be the nodes and edges of a **directed** graph. How many neighbors does node *a* have?

Question 9

1 pts

True or False: In an undirected graph with n nodes, it is possible for there to be an infinite number of **simple paths** between any two nodes.

- True
- False

Question 10

1 pts

Suppose *p* is a shortest path between two nodes *u* and *v* in an undirected graph. True or False: *p* must be a simple path.

- True
- False

Question 11

1 pts

Suppose a directed graph has 42 nodes and 41 edges. Suppose *u* is one of the graph's nodes. What is the greatest possible number of nodes which are reachable from *u*?

Question 12

1 pts

Suppose an undirected graph with 5 nodes has only 3 edges. What is the smallest number of connected components that this graph can have?

Question 13

1 pts

Suppose a given undirected graph has 100 nodes. True or False: if every node in the graph has degree 50 or larger, then the graph must be connected.

- True
- False

Question 14

1 pts

Suppose G is an undirected graph with |V| nodes and |E| edges represented in the computer using a dictionary of sets representation. What is the best possible average case time complexity of any algorithm for finding the node with the largest degree?

- Theta(E)

- Theta(1)
- Theta(V)
- Theta(V^2)
- Theta(V + E)

Question 15

1 pts

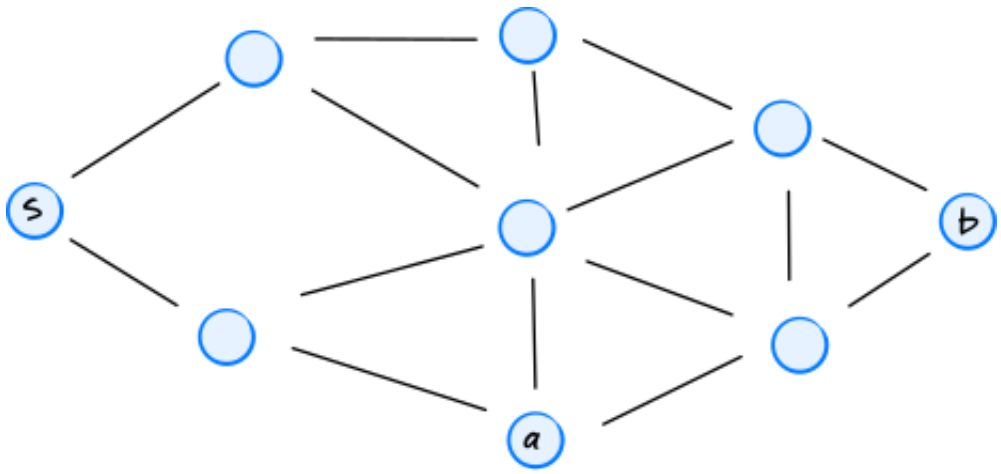
Let A be the adjacency matrix of a graph with n nodes, and let $\vec{e} = (1, 1, \ldots, 1)^\intercal$ be the vector of n ones. The quantity $\vec{e}^\intercal A\vec{e}$ is a number -- what does this number equal?

- The number of connected components
- The number of nodes in the graph
- The number of edges in the graph
- Twice the number of edges in the graph

Question 16

1 pts

Consider running a BFS on the graph below with node s as the source.



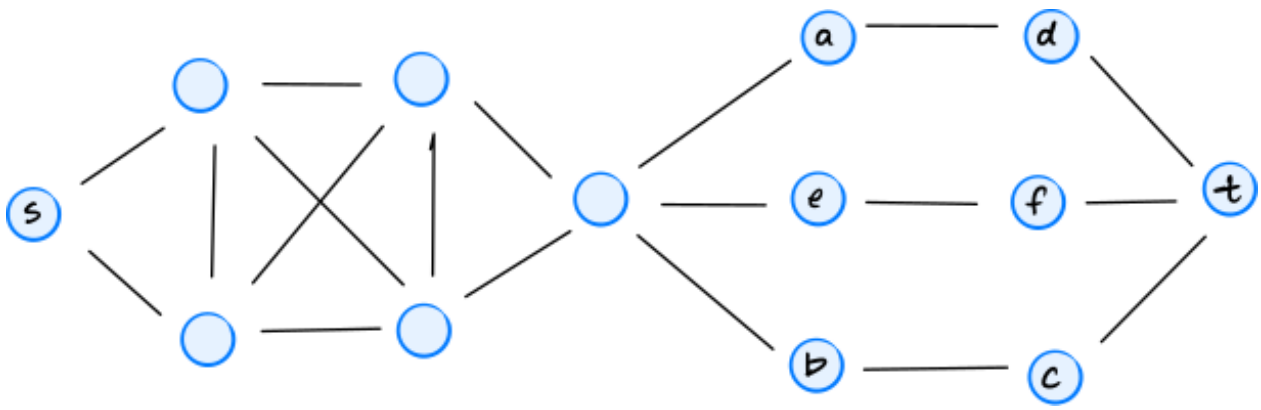
True or False: It is possible for the queue to contain nodes a and b at the same time.

- True
- False

Question 17

1 pts

Consider running a **breadth** first search on the graph below, using node s as the source and the convention that a node's neighbors are produced in ascending order by label. What is the BFS predecessor of node t?

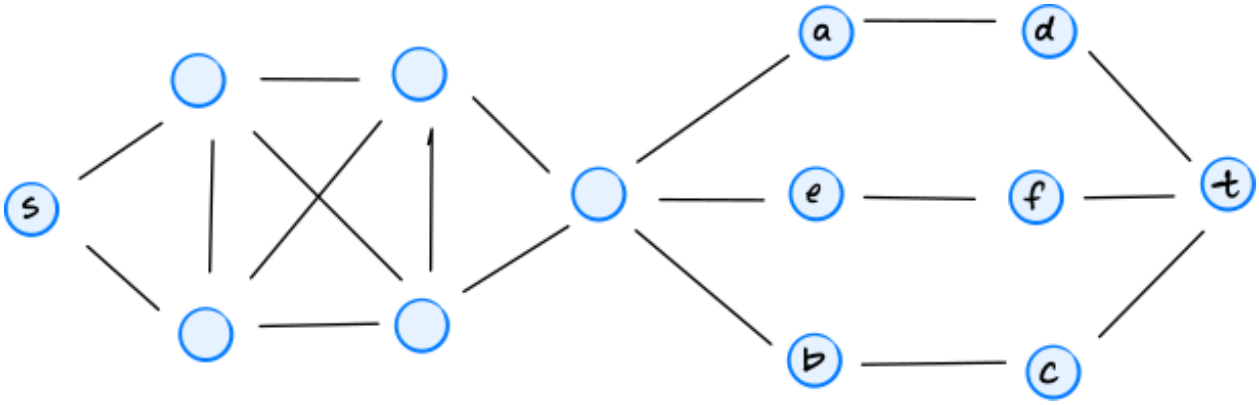


- c
- a
- f
- d

Question 18

1 pts

Consider running a **depth** first search on the graph below, using node s as the source and the convention that a node's neighbors are produced in ascending order by label. What is the DFS predecessor of node t?



- c
- a
- d
- f

Question 19

1 pts

Let *p* be a shortest path between two nodes *u* and *v* in a graph, and suppose that this path passes through a node *w*. True or False: the part of *p* from *u* to *w* must be a shortest path from *u* to *w*.

- True

- False

Question 20

1 pts

Suppose a **depth** first search is run on an **undirected** graph. True or False: it is possible that, at any moment in time, there is a mode marked as visited which has a neighbor node that is marked as pending.

- True
- False

Question 21

1 pts

Consider the following modification of BFS which has an added print statement.

```
from collections import deque

def bfs(graph, source):
    """Start a BFS at `source`."""
    status = {node: 'undiscovered' for node in graph.nodes}

    status[source] = 'pending'
    pending = deque([source])

    # while there are still pending nodes
    while pending:
        u = pending.popleft()
        for v in graph.neighbors(u):
            # explore edge (u,v)
            print(u,v)
            if status[v] == 'undiscovered':
                status[v] = 'pending'
                # append to right
                pending.append(v)
        status[u] = 'visited'
```

Suppose it is run on a connected, undirected graph with 12 nodes and 21 edges. Exactly how many times will something be printed?

Question 22

1 pts

Suppose G is a directed acyclic graph with n nodes. Nodes u, v, and w are three of the nodes in the graph. Node w is a successor of node v, and node v is a successor of node u. Suppose a DFS is run on the graph, starting at some unknown source node. True or False: the finish time of node w must be less than the finish time of node u.

- True
- False

Question 23

1 pts

Let G be a directed graph with with |V| nodes and |E| edges. Suppose someone comes to you with an ordering of the nodes and claims that this is a valid topological sort of the graph. What is the best possible worst case time complexity of an algorithm which checks to see if this supposed ordering is indeed a valid topological sort? You may assume that your graph is represented as an adjacency matrix.

- Theta(V * E)
- Theta(V log V)
- Theta(V)
- Theta(E)
- Theta(V + E)

Quiz saved at 4:40pm

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